

Claude Code Build Brief GODMODE Order Flow EA (cTrader + MT5)

CLAUDE CODE BUILD BRIEF — GODMODE ORDER FLOW EA (cTrader + MT5)

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CLAUDE CODE BUILD BRIEF — GODMODE ORDER FLOW EA (cTrader + MT5)

Project codename: GODMODE_OFEA **Target platforms:** cTrader (cAlgo C#) + MetaTrader 5 (MQL5) — same logic, two builds, same parameters **Target win rate:** 72-78% on A+ filtered setups | Base expectancy +1.6R/trade **Operator:** Bokgabane (SAST timezone, UTC+2) **Source authority:** GODMODE Order Flow Mastery skill v3.0 + the 72% Win-Rate System filter spec + Fabio Valentini AMT methodology

SECTION 0 — HOW TO READ THIS BRIEF

This is a **single, indivisible build specification**. Do not split it into a “later” pile. Every section is a contract — Claude Code must implement what it says, in the order it says, in both languages. Where MQL5 and C# diverge, the brief gives both.

The EA is **not just an executor** — it is also a **visual confluence dashboard**. As confirmations fire one by one, the chart must light up with notifications, symbols, lines, and dashboard rows so Bokgabane can choose to take the trade manually OR let the EA fire automatically. Both modes coexist.

Two operating modes (parameter-switchable): - **MODE_AUTO** — EA fires when all gates pass - **MODE_MANUAL** — EA only displays notifications/symbols/lines; Bokgabane decides

Default = **MODE_MANUAL** **for the first 30 days. After confidence is built, flip to** **MODE_AUTO** .

SECTION 1 — THE TRADING THESIS THE EA MUST EMBODY

The EA is the implementation of **one specific philosophy** — capture this thesis verbatim in the file header comment of both builds:

*Markets spend 70-80% of their time in balance and 20-30% in imbalance. The edge is not predicting which state is next — it is **reacting** to state-change signatures left by institutions: absorption, stacked imbalances, CVD divergence, and aggression prints, at high-probability volume-profile locations, during high-probability sessions, with HTF alignment.*

*Every trade must satisfy the OFT Edge Formula: **Edge = Location × Signal × Context**. All three present → A+ trade. One missing → half size or skip. Two missing → no trade.*

*Operator priority: **capital protection > expectancy > win rate**. The EA refuses entries before it accepts them.*

This thesis is enforced through 5 gates and 8 checklist items, defined below. The EA never fires unless **all 8 checklist items return TRUE** (in **MODE_AUTO**) or notifies on each one as it confirms (in **MODE_MANUAL**).

SECTION 2 — THE CONFLUENCE STACK (THE 5 FILTER RULES)

These are the 5 filters that take the base 48-54% win rate to 72%+. **The EA must enforce all 5 simultaneously**. Each filter is a hard gate.

#	Filter	Rule	Failure action
F1	TRIPLE CONFLUENCE MINIMUM	Require ALL three: VP Level + Footprint Signal (Absorption / Stacked Imbalance / CVD Divergence) + Kill Zone context. Single or double = SKIP.	Block entry, notify “F1 fail: only N/3 confluences”
F2	KILL ZONE DISCIPLINE	Trade ONLY during London Main (10:30-14:00 SAST) or NY Main (17:30-21:00 SAST). 80%+ of A+ setups occur here.	Block entry, notify “F2 fail: outside kill zone”
F3	HTF ALIGNMENT GATE	H4 AND Daily must align with trade direction. Counter-HTF entries cut win rate 20-25%.	Block entry OR halve size if <code>HalfSize_OnHTFConflict=true</code>
F4	CVD CONFIRMATION MANDATORY	Every trade must have CVD confirming direction (rising for long, falling for short) OR showing divergence that supports the bias.	Block entry, notify “F4 fail: CVD against trade”
F5	VOLUME PROFILE STATE CHECK	Balanced (D-Profile) → Model 2 only. Imbalanced (Thin/P/b) → Model 1 only. Never apply M1 in balance or M2 in trend.	Switch model OR skip if state is <code>UNKNOWN</code>

Implementation rule: Every gate must produce a structured log line *and* a chart notification. Bokgabane must be able to look at the chart and *see* which gate is currently TRUE/FALSE in real time.

SECTION 3 — THE 8-POINT ENTRY CHECKLIST (ON-CHART DASHBOARD)

The EA must render a persistent dashboard panel (top-right corner) showing all 8 checklist items with green/red dots. As price moves, dots flip live.

GODMODE OF EA — EURUSD M5 — MODE: MANUAL

① Market State Identified ● BALANCED

② Kill Zone Active ● LDN MAIN

③ HTF Bias Aligned ● BULLISH

④ Price at VP Level ● VAL

⑤ CVD Confirms Direction ● BULLISH

⑥ Footprint Signal Confirmed ○ WAITING

⑦ SL Level Defined ● 1.0832

⑧ R:R ≥ 2:1 ● 3.4:1

CONFLUENCE: 7/8 · PRIORITY: P2 · WAIT

CVD: +14,820 · Δ: -312 · VOL Z: 1.8

POC 1.0867 VAH 1.0892 VAL 1.0830

Daily DD: -0.4% · Trades today: 2

The dashboard is the **single source of truth for the operator**. Both MQL5 and cAlgo must render it identically.

SECTION 4 — THE NOTIFICATION CASCADE (CORE OPERATOR FEATURE)

This is the feature that lets Bokgabane *watch the EA think*. As each confluence layer fires, a sequenced notification appears on the chart in real time. The operator can take the trade manually at any point, or wait for the full cascade.

Notification protocol — fire each in order, one per gate, only when it transitions FALSE→TRUE:

Tag	Meaning	Visual on chart	Sound	Push (mobile)
N-A	Volume Profile signal — price reached VAH/VAL/POC/LVN/HVN	Yellow horizontal dashed line at level + label “VAL TOUCH”	single beep	yes
N-B	Triple confluence achieved — VP + Footprint + Kill Zone all true	Green/red triangle below/above bar + label “TRIPLE OK”	double beep	yes
N-C	Kill zone active (entered London Main or NY Main)	Vertical dotted line at session start + shaded session background	none	no
N-D	HTF alignment confirmed — H4 + Daily agree with bias	“HTF✓” badge in dashboard + green border flash	none	no
N-E	CVD confirmation — CVD rising/falling/diverging in trade direction	“CVD↑” or “CVD↓” arrow next to price + dashboard CVD row turns green	single beep	yes
N-F	Volume Profile state classified (Balanced or Imbalanced)	“BAL” or “IMB” tag in dashboard	none	no
N-G	Footprint signal fired — Absorption / Stacked Imbalance / Unfinished Auction	Star symbol with label “★ABS” / “★SI3” / “★UA” at price level	double beep	yes
N-H	Aggression trigger — large volume Z-score spike at level	Big arrow at candle close + “AGGRESSION ↑” or “↓” + size of print	triple beep + alert	yes (HIGH priority)
N-I	All 8 checklist items pass — A+ entry ready	Full screen overlay flash (green/red) + “ENTER LONG” / “ENTER SHORT” + suggested SL/TP	bell + popup	yes (URGENT)
N-J	Trade fired (in AUTO) or skipped (in MANUAL after timeout)	Entry arrow + SL/TP lines drawn	confirmation chime	yes
N-K	Position management event — partial close at 1R, BE move, trail tighten, POC exit	small label at bar showing event	none	yes
N-L	Kill switch — daily DD exceeded, spread blowout, news blackout, session ending	Red banner across top of chart	alarm	yes (URGENT)

Implementation: - MQL5: use `Alert()` for sound + popup, `SendNotification()` for push, `ObjectCreate()` for chart objects (OBJ_LABEL, OBJ_HLINE, OBJ_VLINE, OBJ_ARROW, OBJ_TEXT, OBJ_RECTANGLE) - cAlgo: use `Notifications.PlaySound()`, `Notifications.SendEmail()`, `Notifications.SendSms()` if configured, `Chart.DrawText()`, `Chart.DrawHorizontalLine()`, `Chart.DrawVerticalLine()`, `Chart.DrawIcon()`, `Chart.DrawRectangle()`

Notification rate-limiting: Each tag may only fire once per bar to prevent spam. State changes are tracked in a `Dictionary<string, bool>` and only edge transitions trigger the notification.

SECTION 5 — THE EXECUTION FLOW (IDENTICAL ACROSS BOTH PLATFORMS)

This is the bar-close cycle. **Implement the gates in this exact order** — do not reorder. If any gate fails, the cycle terminates and the dashboard updates the failing item to red.

```

ON_BAR_CLOSE (MQL5: OnTick + new-bar guard | cAlgo: OnBar)

├─ GATE 0 — KILL SWITCHES → fail: notify N-L, halt
│  └─ Daily DD ≥ MaxDDPct?
│  └─ Spread > NormalSpread × SpreadMult?
│  └─ News blackout active? (manual JSON)
│  └─ Outside trading window entirely?
├─ GATE 1 — F2 SESSION (Filter Rule 2) → fail: notify, skip new entry
│  └─ Compute SAST hour from server time + offset
│  └─ Determine: Asian / LDN-Open / LDN-Main / NY-Open / NY-Main / After
│  └─ Set ActiveModel: LDN-Main → M2 | NY-Main → M1 | LDN-Open → M1 selective
│  └─ Fire N-C on session entry
├─ GATE 2 — F5 STATE (Filter Rule 5) → fail: skip
│  └─ Compute today's profile shape (D / P / b / thin)
│  │   └─ using last N bars and bin distribution skewness
│  └─ Compute open relative to prior VA (inside/outside)
│  └─ Set MarketState: BALANCED | IMBALANCED | UNKNOWN
│  └─ Cross-check ActiveModel against MarketState — mismatch → SKIP
│  └─ Fire N-F
├─ GATE 3 — F1.A LOCATION → fail: skip
│  └─ Compute POC, VAH, VAL via 50-bin VP over VPLength bars
│  └─ Detect LVN bins (vol < POC_vol × 0.20) and HVN bins (vol > POC_vol × 0.70)
│  └─ Test: |close - level| < ATR(14) × LocTolATR for any key level
│  └─ Record which level: VAL / VAH / POC / LVN / HVN
│  └─ Fire N-A on first touch per bar
├─ GATE 4 — F3 HTF ALIGNMENT → fail: half-size or skip
│  └─ Read H4 close vs H4 EMA(20): above = bull, below = bear
│  └─ Read D1 close vs D1 EMA(50): above = bull, below = bear
│  └─ Both must agree with intended trade direction
│  └─ If half-size mode: continue with 0.5x risk
│  └─ Fire N-D on alignment
├─ GATE 5 — F4 CVD CONFIRMATION → fail: skip
│  └─ Aggregate ticks since last bar close → bar_delta
│  └─ Update CVD = CVD + bar_delta
│  └─ Compute CVD slope (last 5 bars) and divergence vs price
│  └─ Long bias: require CVD rising OR bullish divergence
│  └─ Short bias: require CVD falling OR bearish divergence
│  └─ Fire N-E
├─ GATE 6 — F1.B FOOTPRINT SIGNAL → fail: skip
│  └─ Compute volume Z-score over Lookback bars (default 20)
│  └─ Compute delta Z-score over same window
│  └─ Test bullish absorption: delta < 0 AND vol_z ≥ 1.5 AND close ≥ open
│  └─ Test bearish absorption: delta > 0 AND vol_z ≥ 1.5 AND close ≤ open
│  └─ Test stacked imbalance: 3+ consecutive same-direction footprint rows
│  └─ Test unfinished auction: 0-volume row at bar extreme
│  └─ Compute confidence stars (1-5) per absorption confidence formula
│  └─ Require ≥ MinStars (default 3) to pass
│  └─ Fire N-G with star count
├─ GATE 7 — TRIGGER CONFIRMATION → fail: wait next bar
│  └─ M5 CHOCH in entry direction? (last 5 bars structural shift)
│  └─ Aggression candle: vol_z ≥ AggrZ (default 2.0) AND delta sign matches
│  └─ Fire N-H with print size
├─ GATE 8 — SCORE & SIZE → score < 3: skip | 3-4: half | 5: full
│  └─ Compute trade score (0-5):
│  │   └─ +1 VP level | +1 Δ confirms | +1 CVD div confirms
│  │   └─ +1 Absorption present | +1 Session correct
│  └─ Determine Priority tier P1-P5 per the Priority Hierarchy table
│  └─ Compute lots = (Equity × RiskPct/100) / (sl_pips × pip_value)
│  └─ Normalise to broker lot step
│  └─ Fire N-I with full setup summary
├─ EXECUTE (MODE_AUTO only; MODE_MANUAL stops here and waits)
│  └─ EnterLong / EnterShort with magic, label, SL, TP
│  └─ Log to CSV: timestamp, score, signal, levels, sizes
│  └─ Fire N-J
└─ MANAGE EXISTING POSITIONS (every bar, regardless of new-entry path)
   └─ At 1R: close 50%, move SL to BE+1tick, fire N-K
   └─ CVD reverses against position: manual exit, fire N-K
   └─ POC reached (Model 2): close 100% (70% rule), fire N-K
   └─ Trailing: SL = price ± ATR × TrailATR, fire N-K
   └─ Session ending in 5 min: close all, fire N-L + N-K
   └─ Daily DD halt: close all + disable rest of day

```

SECTION 6 — THE 25-SETUP CATALOGUE (BUILT-IN STRATEGY ROUTING)

The EA must recognise and route between all 25 canonical setups from the 72% system PDF. Each setup has its own enable/disable toggle. **Default-enabled = only Priority 1 and 2 setups (the ones with 72%+ win rate).**

Category 1 — Absorption Reversal Setups (Footprint-Based)

#	Setup	Direction	Session	Default	Win Rate
1	Absorption Bottom	LONG	LDN/NY Main	ON	74-78%
2	Absorption Top	SHORT	LDN/NY Main	ON	74-78%
3	CVD Bearish Divergence	SHORT	Any kill zone	ON	72-76%
4	CVD Bullish Divergence	LONG	Any kill zone	ON	72-76%

Category 2 — Volume Profile Setups

#	Setup	Direction	Session	Default	Win Rate
5	VAL Bounce	LONG	LDN Main (M2)	ON	73-77%
6	VAH Fade	SHORT	LDN Main (M2)	ON	73-77%
7	POC Return	BOTH	Balanced	ON	68-72%
8	LVN Acceleration Long	LONG	NY Main (M1)	ON	70-74%
9	LVN Acceleration Short	SHORT	NY Main (M1)	ON	70-74%
10	HVN Rejection	BOTH	LDN/NY Main	OFF (B-grade)	65-69%

Category 3 — Stacked Imbalance Setups

#	Setup	Direction	Session	Default	Win Rate
11	Stacked Bull at VP + CVD	LONG	NY Main	ON	73-77%
12	Stacked Bear at VP + CVD	SHORT	NY Main	ON	73-77%
13	Pullback to Stacked Imbalance Zone	WITH TREND	NY Main	ON	70-74%

Category 4 — Wyckoff 2.0 Setups

#	Setup	Direction	Session	Default	Win Rate
14	Spring Entry (false break + CVD divergence)	LONG	LDN Open 09:00-10:30	ON	75-80%
15	Upthrust (UT/UTAD)	SHORT	LDN/NY	ON	74-78%
16	SOS Breakout (post-Spring)	LONG	After LDN Open	ON	70-74%
17	LPSY Short	SHORT	NY Main	ON	68-73%

Category 5 — ICT/SMC + Order Flow Fusion

#	Setup	Direction	Session	Default	Win Rate
18	Liquidity Sweep + Rejection + CHOCH	BOTH	LDN/NY Kill Zones	ON	74-78%
19	Order Block Return + CVD Confirmation	BOTH	LDN/NY Main	ON	72-76%
20	SMT Divergence (correlated pairs)	BOTH	LDN/NY Open	ON	73-77%
21	Breaker Block Fade	BOTH	NY Main	ON	70-74%
22	ICT AMD Model	BOTH	NY Main 17:30-21:00	ON	75-79%

Category 6 — Microstructure / Auction Theory

#	Setup	Direction	Session	Default	Win Rate
23	Unfinished Auction Fade	FADE	Any kill zone	ON	71-75%
24	Poor High/Low Retest	FADE	Any kill zone	ON	70-74%
25	Iceberg Order Absorption	WITH ICEBERG	NY Main	ON	73-77%

Each setup maps to a `SetupID` enum value 1-25, a `Detect_SetupN()` function, and an `Execute_SetupN()` function. The bar-close loop iterates all enabled setup detectors; any that return TRUE pass through gates 4-8 of Section 5.

SECTION 7 — INPUT PARAMETER SCHEMA (BOTH PLATFORMS, IDENTICAL NAMES)

These parameters must appear in the EA inputs panel exactly as listed. Same names in MQL5 (`input` keyword) and cAlgo (`[Parameter]` attribute).

```

=== MODE & PLATFORM ===
OperatingMode      : enum {MANUAL, AUTO}          default MANUAL
EnableNotifications : bool                      default true
EnablePushAlerts   : bool                      default true
EnableSoundAlerts   : bool                      default true
EnableEmailAlerts   : bool                      default false

=== STRATEGY MASTER ===
EnableModel1_Trend  : bool    default true
EnableModel2_MeanRev : bool    default true
PrimaryTimeframe     : enum {M1,M5,M15,H1}    default M5
HTFBias_Timeframe    : enum {H1,H4,D1}        default H4
HTFConfirm_Timeframe : enum {H4,D1,W1}        default D1
HalfSize_OnHTFConflict : bool    default false // if true, take trade at 0.5x risk

=== SESSION (SAST UTC+2) ===
SAST_OffsetFromBroker : int    default 0        // adjust if broker time != UTC
TradeAsianSession     : bool    default false
TradeLondonOpen       : bool    default false
TradeLondonMain       : bool    default true
TradeNYOpen           : bool    default false
TradeNYMain           : bool    default true
NYOpen_Blackout_Mins  : int     default 20      // skip first 20 min of NY open

=== ORDER FLOW DETECTION ===
DeltaLookback         : int     default 20
VolZThreshold         : double  default 1.5     // absorption volume Z-score min
DeltaZThreshold        : double  default 2.0     // delta Z-score for confidence
AggressionZThreshold   : double  default 2.0     // aggression-print Z-score
MinAbsorptionStars     : int     default 3       // 1-5 confidence
MinStackedImbalanceRows : int    default 3
ImbalanceRatio         : double  default 3.0     // buy_vol > sell_vol * 3.0

=== VOLUME PROFILE ===
VPLength              : int     default 100     // bars for current-day VP
VPBins                : int     default 50
VAValueAreaPct        : double  default 0.70
LVNRatio              : double  default 0.20    // LVN if vol < POC * 0.20
HVNRatio              : double  default 0.70    // HVN if vol > POC * 0.70
LocToATR              : double  default 0.5     // distance to level in ATR

=== RISK MANAGEMENT ===
RiskPct               : double  default 1.0     // % of equity per trade
RiskPct_HalfMode      : double  default 0.5     // half size for B+ setups
MaxRR                 : double  default 6.0
MinRR                 : double  default 2.0
MaxDDPct              : double  default 5.0     // daily drawdown halt
MaxConsecLosses       : int     default 3       // halt after N losses
MaxSpreadMult         : double  default 2.0     // halt if spread > Mult * normal
MagicNumber           : int     default 202604

=== TRADE MANAGEMENT ===
UsePartialClose       : bool    default true
PartialClosePct       : double  default 50.0
PartialCloseAtR       : double  default 1.0
MoveSLToBE_AtR       : double  default 1.0
UseTrailingStop       : bool    default true
TrailATRMult         : double  default 1.0
ExitAtPOC             : bool    default true    // 70% rule for Model 2

=== VISUALISATION ===
ShowDashboard         : bool    default true
ShowVPLevels         : bool    default true
ShowCVDLine          : bool    default true
ShowFootprintMarkers  : bool    default true
ShowSessionShading    : bool    default true
ShowTradeArrows       : bool    default true
DashboardColor_OK     : color    default LimeGreen

```

```
DashboardColor_FAIL      : color default Red
DashboardColor_WAIT      : color default Goldenrod
```

```
=== SETUP TOGGLES (1-25) ===
EnableSetup_01_AbsBot    : bool default true
EnableSetup_02_AbsTop    : bool default true
EnableSetup_03_CVDBear   : bool default true
EnableSetup_04_CVDBull   : bool default true
EnableSetup_05_VALBnc    : bool default true
EnableSetup_06_VAHFade   : bool default true
EnableSetup_07_POCRet    : bool default true
EnableSetup_08_LVNLong   : bool default true
EnableSetup_09_LVNShort  : bool default true
EnableSetup_10_HVNRej    : bool default false
EnableSetup_11_StackBull : bool default true
EnableSetup_12_StackBear : bool default true
EnableSetup_13_PullStack : bool default true
EnableSetup_14_Spring    : bool default true
EnableSetup_15_Upthrust  : bool default true
EnableSetup_16_SOS       : bool default true
EnableSetup_17_LPSY      : bool default true
EnableSetup_18_LiqSweep  : bool default true
EnableSetup_19_OBReturn  : bool default true
EnableSetup_20_SMTDiv    : bool default true
EnableSetup_21_Breaker   : bool default true
EnableSetup_22_AMD       : bool default true
EnableSetup_23_UnfAuc     : bool default true
EnableSetup_24_PoorHL    : bool default true
EnableSetup_25_Iceberg   : bool default true
```

SECTION 8 — MODULAR FILE STRUCTURE (BOTH BUILDS, MIRRORED)

MT5 / MQL5 build:

```
MQL5/Experts/GODMODE_OFEA/
├── GODMODE_OFEA.mqh          # main EA — OnInit/OnTick/OnDeinit/OnTimer
├── Include/
│   ├── OF_Common.mqh        # enums, structs, globals, helpers
│   ├── OF_DeltaEngine.mqh    # tick-delta accumulator + CVD + divergence
│   ├── OF_VolumeProfile.mqh  # POC/VAH/VAL + LVN/HVN bin detection
│   ├── OF_FootprintAnalyzer.mqh # absorption + stacked imbalance + unfinished auction
│   ├── OF_AbsorptionStars.mqh # 1-5 confidence scoring
│   ├── OF_SessionGate.mqh    # SAST window detection + kill zone tags
│   ├── OF_HTFAlignment.mqh   # H4/D1 bias check
│   ├── OF_RiskManager.mqh    # sizing, DD halt, spread guard, consec losses
│   ├── OF_TradeManager.mqh   # partial, BE, trail, POC exit, session close
│   ├── OF_NotificationCenter.mqh # N-A through N-L cascade dispatcher
│   ├── OF_Dashboard.mqh      # 8-row + metrics panel renderer
│   ├── OF_ChartViz.mqh       # VP lines, footprint markers, CVD line, arrows
│   ├── OF_SetupDetectors.mqh # all 25 Detect_SetupN() functions
│   └── OF_Logger.mqh         # CSV journal writer
└── Files/
    └── GODMODE_OFEA_log.csv   # appended on every trade
```

cTrader / cAlgo build:

```
cAlgo/Sources/Robots/GODMODE_OFEA/
├── GODMODE_OFEA.cs          # main Robot — OnStart/OnBar/OnTick/OnStop
├── Modules/
│   ├── OFCommon.cs          # enums, records, helpers
│   ├── DeltaEngine.cs        # Symbol.Ticks aggregator + CVD + divergence
│   ├── VolumeProfile.cs      # POC/VAH/VAL + bin detection
│   ├── FootprintAnalyzer.cs  # absorption + stacked imbalance
│   ├── AbsorptionStars.cs     # confidence scoring
│   ├── SessionGate.cs        # SAST detection
│   ├── HTFAlignment.cs       # H4/D1 bias
│   ├── RiskManager.cs        # sizing + DD + spread + consec losses
│   ├── TradeManager.cs       # partial/BE/trail/POC/session
│   ├── NotificationCenter.cs  # cascade dispatcher
│   ├── Dashboard.cs          # panel via Chart.DrawStaticText
│   ├── ChartViz.cs           # lines/markers/arrows
│   ├── SetupDetectors.cs     # all 25 Detect methods
│   └── TradeLogger.cs         # CSV writer
└── (output: GODMODE_OFEA.algo)
```

Each module exposes the same public function signatures across both languages. When porting, the file-to-file mapping is 1:1 — the porting cost is syntax only, never logic.

SECTION 9 — VISUAL SPECIFICATION (CHART RENDERING)

Render these elements in this exact way. Both platforms must match.

9.1 Volume Profile Levels

- **POC line** — solid, magenta, thickness 2, label “POC ” right-anchored
- **VAH line** — dashed, dodgerblue, thickness 1, label “VAH ”
- **VAL line** — dashed, dodgerblue, thickness 1, label “VAL ”
- **LVN bands** — semi-transparent yellow rectangle 5 pips wide, label “LVN”
- **HVN bands** — semi-transparent green rectangle 5 pips wide, label “HVN”
- **Levels redraw on every bar close** (not every tick — performance)

9.2 CVD Subwindow Indicator

- Separate sub-chart panel below price
- CVD line plotted with colour gradient: green when rising, red when falling
- Divergence arrows: bullish (green up) when price LL + CVD HL; bearish (red down) when price HH + CVD LH
- Y-axis auto-scaled
- Threshold marker line for the rolling 20-bar mean

9.3 Footprint Markers

- **Bullish absorption** — green star ★ below the bar with star count badge (1-5)
- **Bearish absorption** — red star ★ above the bar with star count badge
- **Stacked bull imbalance** — three small blue triangles ▲▲▲ inside the bar
- **Stacked bear imbalance** — three small orange triangles ▼▼▼ inside the bar
- **Unfinished auction** — yellow circle ○ at the unfinished extreme
- **Iceberg detected** — purple diamond ◆ at price level

9.4 Session Shading

- **London Main (10:30-14:00 SAST)** — translucent blue background
- **NY Main (17:30-21:00 SAST)** — translucent orange background
- **NY Open blackout (15:30-15:50)** — translucent grey hatched background
- **Asian / After-market** — no shading (default)

9.5 Trade Lines

- **Entry** — solid line at entry price, label with size and direction
- **SL** — dashed red line, label “SL (-Xpips)”
- **TP** — dashed green line, label “TP (+Xpips, X.XR)”
- **Partial close marker** — small “½” badge at the close bar
- **BE move** — small “BE” badge at the move bar
- **Trail moves** — light grey trail line showing SL evolution

9.6 Dashboard Panel (top-right)

- Rendered via labels (MQL5 OBJ_LABEL with OBJ_PROP_CORNER=CORNER_RIGHT_UPPER, cAlgo Chart.DrawStaticText with VerticalAlignment.Top + HorizontalAlignment.Right)
- Background: semi-transparent dark navy
- Width: 360 px, dynamic height
- Updates every tick (lightweight redraw — change-detect to avoid flicker)
- Shows the 8 checklist items per Section 3 example
- Bottom rows show live metrics: CVD, Δ, Vol Z, POC/VAH/VAL, daily DD, trades today

9.7 Notification Toasts

- Small pop-up rectangle in chart corner, fades out after 4 seconds
- Stack vertically up to 3 visible at once
- Colour-coded: green (confirmation), yellow (waiting), red (failure/halt), blue (informational)

SECTION 10 — DATA FLOW & STATE MANAGEMENT

10.1 Tick-level state (real-time)

```

bid, ask, last
tick_buy_vol_accumulator, tick_sell_vol_accumulator    // reset each bar
spread_pips

```

10.2 Bar-level state (updated on close)

```

bar_delta = tick_buy_vol_accumulator - tick_sell_vol_accumulator
cvd = cvd_prev + bar_delta
volume[i], delta[i] arrays for last N bars
volume_zscore, delta_zscore
poc, vah, val (refreshed per bar)
lvn_levels[], hvn_levels[]
profile_shape: D | P | b | thin
market_state: BALANCED | IMBALANCED | UNKNOWN
htf_bias: BULL | BEAR | NEUTRAL
session: ASIAN | LDN_OPEN | LDN_MAIN | NY_OPEN | NY_MAIN | AFTER
active_model: M1 | M2 | NONE

```

10.3 Trade-level state

```

position_id, entry_price, sl, tp, size, direction
partial_done: bool
be_moved: bool
trail_active: bool

```

10.4 Daily state (resets at session open)

```

daily_start_equity
daily_pnl_pct
trades_today
consec_losses
news_blackout_active

```

10.5 Notification state

```

Dictionary<string, bool> notif_fired_this_bar
Dictionary<string, datetime> notif_last_fired

```

SECTION 11 — SPECIFIC ALGORITHM SPECIFICATIONS

11.1 Bar-Delta Aggregator (the heart of the engine)

SQL5 implementation:

```

// In OnTick: accumulate; on new bar: snapshot and reset.
double g_tick_buy = 0, g_tick_sell = 0;
datetime g_last_bar_time = 0;

void OnTick() {
    MqlTick tick;
    if(!SymbolInfoTick(_Symbol, tick)) return;

    // Direction-flag based aggregation
    if(tick.flags & TICK_FLAG_BUY) g_tick_buy += tick.volume_real;
    if(tick.flags & TICK_FLAG_SELL) g_tick_sell += tick.volume_real;

    datetime cur_bar = iTime(_Symbol, PERIOD_CURRENT, 0);
    if(cur_bar == g_last_bar_time) return;

    // NEW BAR — snapshot delta, reset accumulators, run gates
    double bar_delta = g_tick_buy - g_tick_sell;
    UpdateCVD(bar_delta);
    g_tick_buy = 0; g_tick_sell = 0;
    g_last_bar_time = cur_bar;

    RunAllGates();    // Section 5
}

```

cAlgo implementation:

```

private double _tickBuy = 0, _tickSell = 0;

protected override void OnStart() {
    Symbol.Ticks.Subscribe(OnTickAccumulate);
    // ... rest of init
}

private void OnTickAccumulate(Tick tick) {
    if (tick.TickType == TickType.Ask) _tickBuy += tick.Volume;
    if (tick.TickType == TickType.Bid) _tickSell += tick.Volume;
}

protected override void OnBar() {
    double barDelta = _tickBuy - _tickSell;
    UpdateCVD(barDelta);
    _tickBuy = 0; _tickSell = 0;

    RunAllGates(); // Section 5
}

```

11.2 Volume Profile (POC/VAH/VAL/LVN/HVN)

50-bin distribution over `VPLength` bars. Approximate bar volume as concentrated at the bar's close price (acceptable simplification given M5 granularity). Sweep outward from POC bin, accumulating volume into Value Area until 70% reached.

Pseudocode (apply identically to both languages):

```

high_range = max(high) over VPLength
low_range  = min(low)  over VPLength
bin_size   = (high_range - low_range) / VPBins

for i in 0..VPLength:
    bin = floor((close[i] - low_range) / bin_size)
    vol_bins[bin] += volume[i]

poc_bin = argmax(vol_bins)
poc = low_range + (poc_bin + 0.5) * bin_size

# Value Area expansion
target = sum(vol_bins) * 0.70
acc = vol_bins[poc_bin]
hi = lo = poc_bin
while acc < target:
    up_vol = (hi+1 < VPBins) ? vol_bins[hi+1] : 0
    down_vol = (lo > 0) ? vol_bins[lo-1] : 0
    if up_vol >= down_vol: hi++; acc += up_vol
    else: lo--; acc += down_vol

vah = low_range + (hi + 1) * bin_size
val = low_range + lo * bin_size

# LVN/HVN classification
poc_vol = vol_bins[poc_bin]
for bin in range(VPBins):
    if vol_bins[bin] < poc_vol * LVNRatio: mark as LVN
    if vol_bins[bin] > poc_vol * HVNRatio: mark as HVN

```

11.3 Profile-Shape Classifier (drives F5 state gate)

```

mid_bin    = (highest_filled_bin + lowest_filled_bin) / 2
upper_half = sum(vol_bins[mid_bin..])
lower_half = sum(vol_bins[..mid_bin])
total      = upper_half + lower_half

skewness   = (upper_half - lower_half) / total
peakedness = poc_vol / mean(vol_bins[non-zero])

if abs(skewness) < 0.10 AND peakedness > 2.0: shape = D    # balanced
elif skewness > 0.20:                         shape = P    # bullish trend
elif skewness < -0.20:                        shape = b    # bearish trend
elif peakedness < 1.3:                         shape = THIN # imbalance pending

market_state = (shape == D) ? BALANCED : IMBALANCED

```

11.4 Absorption Star Confidence (1-5)

Per the Absorption Signals spec:

```

stars = 0
if absorption_detected:
    stars = 1
    if vol_zscore >= 2.0:        stars += 1
    if vol_zscore >= 3.0:        stars += 1
    if abs(delta_zscore) >= 2.0: stars += 1
    if relevant_wick_pct >= 0.4: stars += 1 # lower for bull, upper for bear
stars = min(stars, 5)

```

Skip setup if `stars < MinAbsorptionStars`.

11.5 CVD Divergence (drives F4 gate)

```

# Look for last 20 bars
price_HH = highest(high, 20)
price_LL = lowest(low, 20)
cvd_HH   = highest(cvd, 20)
cvd_LL   = lowest(cvd, 20)

# Bullish divergence: price makes lower low, CVD makes higher low → buy bias
bull_div = (low[0] == price_LL) AND (cvd[0] > cvd_LL[1])

# Bearish divergence: price makes higher high, CVD makes lower high → sell bias
bear_div = (high[0] == price_HH) AND (cvd[0] < cvd_HH[1])

```

11.6 SAST Session Detection

```

sast_hour = (broker_hour + SAST_OffsetFromBroker) mod 24
sast_minute = broker_minute
sast_total = sast_hour * 60 + sast_minute

# Define windows in minutes-from-midnight
ASIAN    = [120, 600] # 02:00-10:00
LDN_OPEN = [600, 660] # 10:00-11:00 (also covers 09:00-10:30 for Spring window)
LDN_MAIN  = [660, 930] # 11:00-15:30
NY_OPEN   = [930, 1050] # 15:30-17:30
NY_MAIN   = [1050, 1260] # 17:30-21:00
AFTER     = [1260, 1440] # 21:00-24:00

# NY Open blackout — first 20 min
ny_blackout = sast_total >= 930 AND sast_total < 950

```

11.7 HTF Alignment (drives F3 gate)

```

h4_close = iClose(_Symbol, PERIOD_H4, 0)
h4_ema20 = EMA(20, H4)
h4_bias  = (h4_close > h4_ema20) ? BULL : BEAR

d1_close = iClose(_Symbol, PERIOD_D1, 0)
d1_ema50 = EMA(50, D1)
d1_bias  = (d1_close > d1_ema50) ? BULL : BEAR

aligned = (h4_bias == d1_bias) AND (h4_bias == intended_direction)

```

11.8 Position Sizing (canonical formula)

```

risk_amount = equity * RiskPct / 100
sl_distance = abs(entry - sl)
sl_pips     = sl_distance / pip_size
pip_value_lot = (contract_size * tick_value) / tick_size
lots        = risk_amount / (sl_pips * pip_value_lot)
lots        = clamp(lots, min_lot, max_lot)
lots        = round_to_step(lots, lot_step)

```

SECTION 12 — RISK & KILL-SWITCH RULES (NON-NEGOTIABLE)

These are coded as hard gates and override every other consideration:

1. NO OVERNIGHT POSITIONS
At session end of NY Main (21:00 SAST), close ALL positions, regardless of state.
Cancel pending orders.
2. MAX 1% RISK PER TRADE
RiskPct hard-capped at 2% even if user sets higher. Reject and log.
3. MAX 5% DAILY DRAWDOWN
When $(\text{equity_now} - \text{daily_start_equity}) / \text{daily_start_equity} \leq -\text{MaxDDPct}/100$:
 - CloseAllPositions()
 - Disable new entries for the rest of the day
 - Fire N-L red banner
4. MAX CONSECUTIVE LOSSES = 3
Track via History/PositionsHistory. After 3 consecutive losses → halt for the day.
5. SESSION DISCIPLINE
Refuse new entries outside enabled windows.
6. NEWS BLACKOUT
Read /Files/news_blackout.json (manually maintained or scraped externally).
Format: [{"start": "2026-04-29 14:30", "end": "2026-04-29 14:45", "desc": "NFP"}, ...]
Refuse entries 15 min before through 15 min after each window.
7. MAGIC/LABEL ISOLATION
Every position iteration filters by MagicNumber (MQL5) or Label (cAlgo).
Never touch positions opened by other EAs or by hand.
8. SPREAD GUARD
On each potential entry: $\text{spread_now} > \text{NormalSpread} \times \text{MaxSpreadMult}$ → refuse.
NormalSpread = rolling 100-bar median of spread.

SECTION 13 — LOGGING & JOURNAL

Every trade event writes a CSV row to `GODMODE_OFEA_log.csv`:

```
timestamp_utc, symbol, event_type, setup_id, score, priority,
direction, entry_price, sl, tp, lots, risk_pct,
session, market_state, htf_bias, profile_shape,
poc, vah, val,
cvd_at_entry, bar_delta, vol_z, delta_z, abs_stars,
notif_cascade_completed, mode (AUTO/MANUAL),
result_pnl_pips, result_pnl_pct, result_R, hold_minutes
```

Event types: `ENTRY`, `PARTIAL`, `BE_MOVE`, `TRAIL`, `EXIT_TP`, `EXIT_SL`, `EXIT_POC`, `EXIT_SESSION`, `EXIT_REVERSAL`, `SKIP_F1`, `SKIP_F2`, `SKIP_F3`, `SKIP_F4`, `SKIP_F5`, `KILL_DD`, `KILL_NEWS`, `KILL_SPREAD`.

This file is the operator's truth source for edge-decay tracking.

SECTION 14 — TESTING PROTOCOL (CLAUDE CODE MUST PRODUCE TEST PLAN)

Alongside the EA, generate a test plan document `TESTING.md` containing:

1. **Compilation test** — both files compile zero warnings, zero errors
2. **Visual test checklist** — load on demo chart, verify every notification, every dashboard row, every visual element renders correctly
3. **Backtest protocol (MT5):**
 - 6 months minimum on EURUSD M5
 - Mode: "Every tick based on real ticks"
 - Initial deposit: \$10,000
 - Acceptance: $\text{PF} \geq 1.5$, $\text{Win Rate} \geq 65\%$ on filtered trades, $\text{Max DD} \leq 12\%$, $\text{Recovery Factor} \geq 2.0$
4. **Backtest protocol (cTrader):**
 - 6 months EURUSD M5, tick data
 - Same acceptance metrics
5. **Forward test:** 2 weeks demo at 0.25x size before live
6. **A/B comparison:** run both builds simultaneously on demo accounts; compare fill prices, signal agreement rate, slippage

SECTION 15 — CODE QUALITY REQUIREMENTS

1. **Zero magic numbers in code.** Every threshold is an input parameter.

2. **All functions ≤ 50 lines.** Decompose anything bigger.
3. **Every gate function returns a struct/record:** `{passed: bool, reason: string, value: double}` so the dashboard and logger can render the *why*, not just yes/no.
4. **No global state outside the engine modules.** Inputs and module-level statics only.
5. **Defensive null checks** on every indicator/handle/series read.
6. **Try-catch (cAlgo) / GetLastError checks (MQL5)** on every order operation. Failed order → log + notify N-L, never silent.
7. **Comments at the top of every module** describing: purpose, public API, dependencies, source authority (which section of this brief drives it).
8. **Compilation flags:**
 - MQL5: `#property strict`, treat warnings as errors
 - cAlgo: `[Robot(AccessRights = AccessRights.FileSystem)]` for log writing only — no internet access
9. **No platform-specific shortcuts that break the mirror.** If MQL5 has a feature C# doesn't, simulate it in C# rather than diverging behaviour.
10. **Naming:** `PascalCase` for types and public methods, `camelCase` for locals (C#); `PascalCase` for functions, `g_` prefix for globals (MQL5). Module names match across both.

SECTION 16 — DELIVERABLES

When Claude Code finishes, the working directory must contain:

```
GODMODE_OFEA/
├── README.md                # operator manual: install, configure, run
├── TESTING.md               # test plan per Section 14
├── CHANGELOG.md             # build version, dependencies, source authority
├── mt5/
│   └── GODMODE_OFEA/        # MQL5 build (Section 8)
├── ctrader/
│   └── GODMODE_OFEA/        # cAlgo build (Section 8)
├── docs/
│   ├── architecture.md      # how the gates compose
│   ├── confluence_examples.md # 5 worked examples of notification cascades
│   └── parameter_tuning.md   # which params to optimise, in what order
└── samples/
    ├── news_blackout.json    # example schema
    └── GODMODE_OFEA_log.sample.csv # sample journal
```

SECTION 17 — BUILD ORDER (CLAUDE CODE EXECUTION SEQUENCE)

Build in this order. Do NOT jump ahead — each phase depends on the previous.

PHASE 1 — Scaffold & Common (both platforms in parallel) 1. Create directory structure (Section 8) 2. Build OF_Common / OFCommon (enums, structs, helpers) 3. Build OF_Logger / TradeLogger (CSV writer) 4. Compile-check empty shells

PHASE 2 — Engine Core 5. DeltaEngine (tick aggregator + CVD + divergence) 6. VolumeProfile (POC/VAH/VAL + LVN/HVN + shape classifier) 7. FootprintAnalyzer + AbsorptionStars 8. SessionGate (SAST mapping) 9. HTFAlignment

PHASE 3 — Risk & Trade 10. RiskManager (sizing, DD halt, spread guard, consec losses, news blackout) 11. TradeManager (partial, BE, trail, POC, session close)

PHASE 4 — Decision Layer 12. SetupDetectors — implement all 25 Detect_SetupN() 13. Main bar-close pipeline (Section 5) — wire gates 0-8 in order

PHASE 5 — Operator Surface 14. NotificationCenter (N-A through N-L cascade) 15. ChartViz (VP lines, footprint markers, CVD subwindow, session shading, trade lines) 16. Dashboard (8-row panel + metrics)

PHASE 6 — Integration 17. Wire NotificationCenter into every gate of the bar-close pipeline 18. Wire Dashboard refresh on tick + bar 19. Test MODE_MANUAL: notifications fire, EA does not enter 20. Test MODE_AUTO: full cascade fires entry

PHASE 7 — Documentation & Test Plan 21. README.md with install/config steps for both platforms 22. TESTING.md with backtest protocol 23. docs/* files 24. samples/*

PHASE 8 — Compile & Smoke Test 25. Compile MT5 build in MetaEditor — zero warnings 26. Compile cTrader build in cAlgo IDE — zero warnings 27. Load both on demo charts, walk through one bar of each session, verify dashboard + notifications

SECTION 18 — OPERATOR'S USER STORY (THE NORTH STAR)

When Bokgabane opens the chart, here is what must happen:

*"I attach the EA to EURUSD M5. The dashboard appears top-right showing 8 dots. The chart now has a magenta POC line, blue VAH/VAL dashed lines, yellow LVN bands, and the London Main session is shaded translucent blue because we're inside it. A horizontal yellow line appears at VAL — the EA chimes once and a small toast appears: **N-A: VAL touch at 1.0830. Dot ④***

flips green.

A few bars later, the EA detects a profile shape of D and tags BAL in the dashboard. Dot ① flips green. The H4 EMA confirms bullish, D1 EMA confirms bullish — dot ③ flips green and a small badge “HTF✓” appears.

CVD has been rising for five bars. Dot ⑤ flips green and a soft “CVD↑” arrow appears next to the price. **N-E** chimes.

Now a strong bar closes with negative delta but a bullish body and the volume Z-score hits 2.4. A green ★★★★★ star appears below the bar. **N-G chimes twice**: ‘Bullish Absorption — 4-star’. Dot ⑥ flips green.

Confluence count is 7/8. The EA computes SL at 1.0822 (1-tick below the absorption candle low) and TP at the POC 1.0867. R:R = 4.4:1. Dots ⑦ and ⑧ flip green.

All 8 are green. **N-I chimes a bell** and a green flash overlays the chart with a popup: ‘A+ ENTRY READY — LONG EURUSD — Score 5/5 — P1 — Full Size’.

Because I’m in MANUAL mode, nothing fires. I look at the chart. I see all the confluences clearly. I’m satisfied with the setup. I click Buy on my platform. The EA continues to manage the position from there — partial close at 1R, BE move, trail to POC, exit at POC.

If I’d been in AUTO mode, the EA would have entered the moment **N-I** fired. Either way, the thinking is visible at every stage.”

This user story is the acceptance test. If the EA doesn’t behave this way end-to-end, it isn’t done.

SECTION 19 — WHAT NOT TO DO (HARD PROHIBITIONS)

1. **Do not use TradingView, Pine Script, or webhooks.** Two-platform build only: cTrader + MT5.
2. **Do not invent parameters not listed in Section 7.** If a value is needed, propose it as a new input — don’t hardcode.
3. **Do not use `sleep()` or busy-wait loops.** All timing is event-driven (OnTick / OnBar / OnTimer).
4. **Do not place stops at “obvious” swing highs/lows.** Stops go 1-2 ticks beyond the aggression candle, per Fabio’s rule.
5. **Do not exit beyond the POC in Model 2.** The 70% rule is mandatory — full exit at POC.
6. **Do not deviate from the 8-gate order.** F2 → F5 → location → F3 → F4 → footprint → trigger → score. This sequence is the edge.
7. **Do not write platform-divergent logic.** Both builds compute the same numbers from the same rules. Different syntax, identical behaviour.
8. **Do not skip the dashboard or notifications.** They are not optional decoration — they are the operator’s interface.
9. **Do not allow new entries while a position is open** unless the input `AllowPyramiding=true` (default false).
10. **Do not implement features beyond this spec.** No “while we’re at it” additions. If something is missing, document it in CHANGELOG.md as a Phase-2 candidate, then stop.

SECTION 20 — FINAL STANDING ORDER

Build this EA exactly as specified. Mirror the logic across MQL5 and cAlgo. Render the dashboard, fire the notifications, enforce the 5 filters, route the 25 setups, log every event. The operator must be able to see the EA think, gate by gate, and choose to follow it or override it. Capital protection is the first principle. The 72%+ win rate is a consequence of discipline, not a target to chase.

Now build it.

SECTION 21 — WIN-RATE CEILING & HONEST TRADEOFFS (READ BEFORE §22)

This section exists because the operator asked about pushing win rate to 80%+. The honest engineering answer is documented here so future tuning decisions are anchored to reality, not marketing.

21.1 The mathematical ceiling

System tier	Win rate	Avg win	Avg loss	Expectancy/trade	Trades/day
Base OFT (no filters)	48-54%	2.8R	1.0R	+0.94R	10-15
72% filtered (5 gates)	72-78%	3.2R	1.0R	+1.60R	5-10
Realistic ceiling with marginal gains (§22)	75-78%	3.2-3.4R	0.9R	+1.85R	5-10
Theoretical 80%+ on this style	not sustained	—	—	—	not at this volume

Why 80%+ is not a realistic operating target for a discretionary order-flow EA at 5-15 trades/day:

1. Renaissance Medallion, the most successful quant fund in history, runs around 71% on sub-second timeframes with infrastructure no retail trader can replicate. A retail M5 EA cannot exceed that on a sustained basis.
2. Public, audited records from top discretionary order-flow traders (Fabio Valentini included) cluster at 55-65%. The 72%+ target the brief already sets is at the upper edge of what disciplined human execution achieves.
3. Sustained 80%+ win rates almost always conceal one of three statistical traps (see §21.3) that destroy net profit even as the win rate looks good.

21.2 Why each additional 5% costs more than the last

The relationship between filter strictness and trade count is exponential, not linear:

Filter intensity	Approx. setups passed per day	Win rate
Base	15-20	48-54%
5 gates (current brief)	5-10	72-78%
5 gates + marginal stack (§22)	5-10	75-78%
6+ gates with strict tiering	1-3	80-82%
7+ gates with multi-symbol weighting	<1 (most days zero)	84%+

To preserve 5-15 trades/day at 80%+ win rate, the filter would have to *both* tighten *and* widen the universe — which the brief addresses in §23 by adding parallel symbol scanning, not by relaxing filters.

21.3 The three traps that fake an 80% win rate

The EA must be coded so it cannot accidentally fall into any of these:

Trap	What it looks like	What it actually is	Brief enforcement
Tiny-R trap	80% wins of 0.3R, 20% losses of 1R	Negative expectancy: -0.04R/trade	§7 input <code>MinRR ≥ 2.0</code> is hard-enforced; reject trades below this regardless of win rate
Martingale trap	Win rate looks 90%+ on backtest	One full SL hit wipes weeks of “wins” because position size escalates after losses	§7 has no martingale parameter; size is always <code>Equity × RiskPct</code> from the <i>current</i> equity, not adjusted for recent results
Survivorship trap	80% on a 30-trade sample	Statistical noise — confidence interval at n=30 is ±15%	§22.8 mandates a 200-trade rolling sample before claiming any win-rate metric

21.4 What “win rate” should and shouldn’t drive

- **Drive optimisation by:** Expectancy/trade, Profit Factor, R-multiple distribution, max drawdown, recovery factor.
- **Do not drive optimisation by:** Win rate alone. A 60% / +1.5R system beats an 80% / +0.6R system in real money.

The EA’s `TESTING.md` acceptance criteria (§14) reflect this — they target Profit Factor ≥ 1.5 and Recovery Factor ≥ 2.0, not a win-rate floor.

SECTION 22 — THE MARGINAL-GAIN STACK (REALISTIC PATH FROM ~72% TO ~77-78%)

These eight refinements add to the existing 5 gates without removing trades. Each has an **expected marginal lift** based on backtested data from the source authorities. Lifts compound *partially* (not additively) because the filters overlap. Realistic stacked

lift: **+4 to +6 percentage points** on win rate.

Add these as additional input parameters and additional gates in the bar-close pipeline. They run *after* the existing 5 gates, before final scoring.

22.1 Refinement M1 — Liquidity-Sweep Confirmation Layer

Source authority: ICT/SMC integration in skill ref-04, Wyckoff Spring protocol ref-05, 72% PDF setup #18. **Lift:** +1.0 to +1.5 pp.
Require — only on *reversal* setups (Setups 1-6, 14-22) — that within the previous 10 bars price made a sweep of an obvious BSL (buy-side liquidity = recent swing high) or SSL (sell-side liquidity = recent swing low) and *closed back* on the opposite side. Sweeps without close-back rejection do not count.

```
input bool    M1_RequireLiquiditySweep = true;
input int     M1_SweepLookbackBars    = 10;
input int     M1_SwingDetectionBars    = 20;
```

Implementation: detect swing high/low over `M1_SwingDetectionBars`. Within the last `M1_SweepLookbackBars`, did `high > swing_high` then `close < swing_high` ? (mirror for low). If yes → reversal setup is liquidity-confirmed.

22.2 Refinement M2 — Two-Touch Rule on VP Levels

Source authority: Fabio Valentini “Wait for the Second Wave / Retest” rule, Strategy Guide §07. **Lift:** +1.0 to +1.5 pp.
The first touch of VAH/VAL/POC is a probe — not an entry. Require that price has tested the level *at least twice* within the lookback window before allowing entry. The second-touch failure is statistically twice as likely to hold.

```
input bool    M2_RequireSecondTouch    = true;
input int     M2_TouchLookbackBars      = 30;
input double  M2_TouchToIPips           = 3.0;
```

Implementation: maintain a `touch_count[level]` counter per level. Increment on each bar where `|close - level| < M2_TouchToIPips * pip_size`. Require `touch_count >= 2` before allowing an entry at that level.

22.3 Refinement M3 — Volatility Regime Filter

Source authority: Fabio §11 “Volatility Regime Change” warning on edge decay. **Lift:** +0.5 to +1.0 pp.
Skip trading entirely when ATR is in an abnormal regime — neither too quiet (no follow-through) nor too volatile (algos hunt stops aggressively). The sweet spot is 0.7x to 1.5x the rolling 50-day ATR average.

```
input bool    M3_UseATRRegimeFilter      = true;
input double  M3_MinATRRatio             = 0.70;
input double  M3_MaxATRRatio             = 1.50;
input int     M3_ATRBaselineDays         = 50;
```

Implementation: compute `atr_ratio = ATR(14, current TF) / mean(ATR(14, daily) over M3_ATRBaselineDays)`. Block entries if outside `[M3_MinATRRatio, M3_MaxATRRatio]`. Notify N-L when blocked.

22.4 Refinement M4 — News Proximity Hard-Block (Tiered)

Source authority: Fabio §11, 72% PDF risk rules. **Lift:** +0.5 to +1.0 pp (avoiding the random-walk losses around news).
The current spec already has news blackout. Tighten it into three tiers based on news impact severity:

Impact	Block window before	Block window after
HIGH (NFP, FOMC, CPI)	30 min	30 min
MEDIUM (PPI, retail sales)	15 min	15 min
LOW	not blocked	not blocked

```
input bool    M4_TieredNewsBlackout      = true;
input int     M4_HighImpact_PreMin       = 30;
input int     M4_HighImpact_PostMin      = 30;
input int     M4_MedImpact_PreMin        = 15;
input int     M4_MedImpact_PostMin       = 15;
```

Update the news JSON schema:

```
[{"start": "2026-04-29 14:30", "end": "2026-04-29 14:30", "impact": "HIGH", "desc": "NFP"}]
```

22.5 Refinement M5 — Dual-CVD (Price-CVD + Symbol-CVD) Convergence

Source authority: Darius FX foundations, ref-07; Mind Math Money 83-min. **Lift:** +1.0 to +1.5 pp.
Currently the EA uses one CVD (the trading symbol). Add a second CVD from a *correlated* symbol (e.g. DXY for EURUSD, US100 for NASDAQ futures) and require *both* CVDs to agree.

If both CVDs confirm direction → standard size. If only the primary CVD confirms (correlated CVD is neutral) → half size. If correlated CVD diverges against the trade → SKIP regardless of primary.

```
input bool    M5_UseCorrelatedCVD      = true;
input string  M5_CorrelatedSymbol      = "DXY"; // user sets per primary symbol
input bool    M5_HalfSizeOnNeutral     = true;
```

Implementation: subscribe to ticks/bars on `M5_CorrelatedSymbol` (MT5: `MarketWatch` , cAlgo: `MarketData.GetSeries`). Run a parallel CVD aggregator. Compare slope/divergence sign at decision time.

22.6 Refinement M6 — Time-of-Day Sub-Window Filter

Source authority: 72% PDF session map, Fabio \$05. **Lift:** +0.5 to +1.0 pp. Within London Main and NY Main, certain sub-windows have measurably higher A+ density. The brief currently treats the whole session as equal. Tier the windows:

Sub-window (SAST)	Tier	Allowed setups
LDN Main 11:00-13:30	TIER A	All M2 setups
LDN Main 13:30-15:30	TIER B	M2 only if confluence ≥ 5/5
NY Main 17:30-19:00	TIER A	All M1 setups
NY Main 19:00-21:00	TIER B	M1 only if confluence ≥ 5/5

```
input bool    M6_UseSubWindowTiering  = true;
input int     M6_TierBMinScore        = 5; // require 5/5 in tier B windows
```

22.7 Refinement M7 — Spread/Slippage Quality Gate

Source authority: Fabio \$09, 72% PDF risk rules. **Lift:** +0.3 to +0.7 pp (mostly via reduced cost drag, which lifts realised win rate).

Beyond the existing spread guard, add a *slippage history* check. Track average slippage on the last 20 fills. If rolling slippage > threshold → refuse new entries for the rest of the session and notify operator that broker conditions deteriorated.

```
input bool    M7_UseSlippageGate      = true;
input double  M7_MaxAvgSlippagePips   = 0.5;
input int     M7_SlippageWindowFills  = 20;
```

Implementation: on each fill, log `requested_price - fill_price` in pips. Maintain rolling average. If exceeded → set `g_slippage_halt = true` for the session.

22.8 Refinement M8 — Adaptive Confluence Tiering (rolling sample)

Source authority: the math in \$21.3 — only trust win-rate metrics on $n \geq 200$. **Lift:** +0.5 to +1.0 pp via avoiding overfitted entries during regime transitions.

Maintain a rolling 200-trade sample of every realised trade (pulled from the journal). For each setup ID 1-25, compute the rolling win rate. If a setup’s rolling win rate drops below 60% for >50 of its own trades → auto-disable that setup until manually re-enabled.

```
input bool    M8_AdaptiveSetupTiering = true;
input int     M8_RollingSampleSize     = 200;
input double  M8_AutoDisableThreshold  = 0.60;
input int     M8_MinSampleBeforeAction = 50;
```

This is the EA’s edge-decay defence. It lets winners run and quietly retires setups that stop working, without the operator having to monitor the journal manually.

22.9 Stacked expected outcome

Layer	Expected lift	Cumulative win rate
Base 5-gate (current brief)	—	~72-78%
+ M1 Liquidity sweep	+1.0-1.5 pp	~73-79%
+ M2 Two-touch rule	+1.0-1.5 pp	~74-80%
+ M3 ATR regime	+0.5-1.0 pp	~74.5-80.5%
+ M4 Tiered news	+0.5-1.0 pp	~75-81%
+ M5 Dual-CVD	+1.0-1.5 pp	~75.5-82%
+ M6 Sub-window tier	+0.5-1.0 pp	~76-82.5%
+ M7 Slippage gate	+0.3-0.7 pp (cost)	~76-82.5%
+ M8 Adaptive tiering	+0.5-1.0 pp	~76.5-83% upper bound

Realistic operating expectation: the lifts overlap — they’re not independent. Real-world stacked lift is **+4 to +6 percentage points**, putting the EA at **~76-78% sustained**, with **occasional rolling 50-trade samples touching 80-82%** during favourable regimes. This is the honest ceiling for this style at this volume.

SECTION 23 — TRADE-COUNT PRESERVATION PROTOCOL (KEEPING 5-15 TRADES/DAY)

Adding more filters reduces trades. To prevent that, **expand the universe horizontally**, not the filter set vertically. The EA must scan multiple symbols simultaneously and route the highest-confluence setups across all of them into the existing trade slot.

23.1 Multi-symbol scanning architecture

```
input bool    MultiSymbol_Enable          = true;
input string  MultiSymbol_List            = "EURUSD,GBPUSD,USDJPY,XAUUSD,US100,US500";
input int     MultiSymbol_MaxConcurrent   = 2;    // max simultaneous open positions
input int     MultiSymbol_MaxPerDay       = 12;    // hard cap across all symbols
input bool    MultiSymbol_CorrelationFilter = true; // skip if correlated to open position
```

Implementation pattern: - The main EA chart runs the dashboard and the operator surface - A **headless worker function** runs the same gate pipeline against each symbol in `MultiSymbol_List` - Each symbol gets its own `SymbolState` struct (CVD, VP, session, scores) - The dispatcher selects the highest-priority setup at any given moment across all symbols and routes it to the trade engine - Open positions on correlated symbols (e.g. EURUSD long + GBPUSD long simultaneously) are blocked by the correlation filter to prevent disguised concentration

23.2 The math of universe expansion

Symbols scanned	A+ setups per symbol per day	Total A+ setups available
1 (current brief)	5-10	5-10
3 majors	5-10 each	8-14 (after correlation dedup)
6 majors + indices	5-10 each	12-18 (after correlation dedup)

So adding the marginal-gain stack (which roughly halves per-symbol setups) while expanding to 3-6 symbols preserves 5-15 trades/day at higher quality.

23.3 Correlation filter spec

Symbol A	Correlated symbols (block simultaneous trades)
EURUSD	GBPUSD, AUDUSD, NZDUSD, EURGBP, DXY (inverse)
USDJPY	USDCHF, US10Y, DXY
XAUUSD	XAGUSD, AUDUSD, DXY (inverse)
US100	US500, US30
US500	US100, US30, VIX (inverse)

If a position is open on a correlated symbol, the second setup is **logged but skipped** with reason `SKIP_CORR` so it shows in the journal but doesn’t increase concentration risk.

23.4 Symbol-rotation discipline

Even with 6 symbols, do not allow the EA to become “the EURUSD EA” because EURUSD happens to fire most often. Cap each symbol at:

```
input int      MultiSymbol_MaxPerSymbol_PerDay = 4;
```

This ensures distribution across the universe and prevents over-concentration in whichever symbol happens to be in a tradeable regime.

23.5 Multi-symbol dashboard upgrade

The \$3 dashboard becomes multi-row when `MultiSymbol_Enable=true` :

GODMODE OF EA — 6 SYMBOLS — MODE: MANUAL									
EURUSD	●●●●●○	7/8	P2	CVD↑	AT VAL	WAIT FP			
GBPUSD	●●●●●○	7/8	P2	CVD↑	CORR-LOCK	BLOCKED			
USDJPY	○●●●●○	4/8	—	—	OUT-LOC	SKIP			
XAUUSD	●●●●●●	8/8	P1	CVD↓	AT VAH	READY *****	ENTER		
US100	●●●●●○	6/8	P3	CVD↑	AT POC	WAIT KZ			
US500	○●●●●○	6/8	P3	CVD↑	AT POC	WAIT KZ			
Daily DD: -0.4% Trades today: 4/12 Open: XAUUSD (L) US100 (L)									

Each row updates live. The notification cascade (N-A through N-L) prefixes each notification with the symbol tag.

23.6 Computation budget

Multi-symbol scanning increases CPU load. The brief enforces: - All multi-symbol computations run on the **bar-close event of each symbol**, not on every tick - Tick-level work (delta accumulation) runs in lightweight per-symbol callbacks - Dashboard refresh is throttled to once every 250 ms regardless of tick rate - Per-symbol VP recomputation runs on a 1-minute timer, not per bar, when the symbol has no current setup developing

23.7 Acceptance test for multi-symbol mode

Add to `TESTING.md` : - 6 months backtest across all symbols in `MultiSymbol_List` - Verify trade count averages 8-12/day (within 5-15 target) - Verify no day exceeds `MultiSymbol_MaxPerDay` - Verify no symbol exceeds `MultiSymbol_MaxPerSymbol_PerDay` - Verify correlation filter blocks the expected pairs - Verify aggregated win rate falls in the **76-78% sustained** range with isolated 50-trade windows reaching 80%+

SECTION 24 — UPDATED STANDING ORDER (REPLACES \$20 AS THE FINAL WORD)

Build this EA exactly as specified across \$1 through \$23. Mirror logic across MQL5 and cAlgo. Render the dashboard, fire the notifications, enforce the 5 base gates, layer in the 8 marginal-gain refinements, scan the multi-symbol universe, route through correlation discipline, log every event.

Target the realistic ceiling: **76-78% sustained win rate, +1.85R/trade expectancy, 5-15 trades/day distributed across 3-6 symbols, with rolling 50-trade samples occasionally touching 80-82% in favourable regimes.**

Refuse to chase 80%+ as a hard target. Win rate is a consequence of discipline; expectancy is the goal; capital protection is the first principle.

Now build it.

Brief authored by GODMODE Order Flow Mastery skill v3.0 synthesis. Source authorities: 72% Win-Rate System spec, Fabio Valentini AMT Strategy Guide, ChartFanatics x DeepCharts encyclopedia, Mind Math Money 83-min OFT course, Wyckoff 2.0 deep-dive, Darius FX foundations. Total source coverage: 69+ documents + 7 reference files.

APPENDIX A — PUSHING THE WIN RATE FROM 72-78% TOWARD 80%+

A.0 — HONEST FRAMING (READ THIS FIRST)

The base brief targets **72-78%** on Priority 1 and 2 setups. That figure is supported in the source material and is achievable. Pushing to **80%+ at the same daily trade volume (5-10 trades/day)** is harder than the literature implies. Here is the honest landscape:

Win-rate band	Difficulty	Realism
72-78%	Achievable with the base 5 filters	Documented and reproducible
78-82%	Requires 4-6 additional micro-filters, more discretion, or post-trade analytics feedback	Realistic for skilled operators after 3-6 months of data
82-85%	Requires shrinking the trade universe — fewer trades, not the same volume	Possible but trades off frequency
85%+	Either the operator is in a recency-favourable regime, or the sample is too small to be reliable, or the win is being paid for with worse R:R	Treat with suspicion

The trade-off no marketing material admits: every additional filter that raises win rate also reduces trade count *or* compresses average R. Holding trade count constant **and** pushing win rate above 80% means you must squeeze the additional edge from the *quality of execution* and *post-trade adaptive learning*, not just from adding more entry filters.

Below are 12 techniques. Each carries an honest **estimated win-rate contribution** and an honest **trade-volume side effect**. Implement them as opt-in inputs — never on by default, always parameterised, always logged so you can prove or disprove the contribution from your own data.

A.1 — THE 12 TECHNIQUES — RANKED BY HONEST EXPECTED VALUE

Tier 1 — Highest-confidence, smallest side effects (implement first)

A.1.1 — Quad Confluence (upgrade F1 from triple to quadruple) - Mechanic: Require VP Level + Footprint Signal + Kill Zone + Wyckoff Phase confirmation (in Phase C/D — Spring or LPS confirmed). Currently F1 is “triple”. Adding a fourth source raises the prior probability of institutional involvement. - **Estimated contribution:** +2 to +3 percentage points - **Trade volume side effect:** -10 to -15% (some Priority 2 setups lose Wyckoff confirmation) - **Implementation:** New input `RequireQuadConfluence: bool default false`. When TRUE, gate F1 also checks `wyckoff_phase ∈ {C, D}`. - **Honesty note:** This is the strongest single filter. The reason it works: Wyckoff phase requires *prior* range development, which means you’re trading *after* institutional positioning is documented, not coincident with it.

A.1.2 — Liquidity Sweep Pre-condition - Mechanic: Before any reversal entry, require evidence that a recent liquidity pool was swept (BSL or SSL). Most A+ reversals occur AFTER a sweep — entries before the sweep pay for the sweep. - **Estimated contribution:** +2 percentage points on reversal setups (Cat 1, 4, 5) - **Trade volume side effect:** -5 to -10% (pure trend continuation setups unaffected) - **Implementation:** New module `LiquidityScanner.cs` / `OF_LiquidityScanner.mqh` that detects swept old highs/lows in the last 50 bars. New input `RequireLiquiditySweep_Reversals: bool default false`. Applies only to reversal-direction setups. - **Honesty note:** Detection is fragile — false positives on noisy ranges. Combine with HTF bias gate to reduce noise.

A.1.3 — Round-Number Confluence Bonus - Mechanic: Major round numbers (00, 50 in last 2 digits of price) act as institutional limit-order pools. Setups occurring within `±3 pips` of a round number have measurably higher conversion. This is a *bonus filter* — passes the standard gates faster, doesn’t replace them. - **Estimated contribution:** +1 to +2 percentage points - **Trade volume side effect:** ~0 (existing setups upgraded, not removed) - **Implementation:** New input `PrioritiseRoundNumbers: bool default true`. When TRUE, setups within tolerance get +1 to score (max stays 5). - **Honesty note:** Effect is well-documented in FX microstructure but smaller than retail traders believe. Don’t over-weight it.

Tier 2 — Solid, but with real volume cost

A.1.4 — DOM Imbalance Confirmation (broker-dependent) - Mechanic: At entry, read top-5 levels of DOM (depth of market). If aggregate bid size > aggregate ask size by 2x in the trade direction, treat as institutional limit defense already in place. - **Estimated contribution:** +2 to +3 percentage points - **Trade volume side effect:** -10% (some setups have no DOM confirmation) - **Implementation:** MT5 `MarketBookGet()` + `OnBookEvent`. cAlgo `MarketDepth.Updated` event. New input `RequireDOMConfirmation: bool default false`. **WARNING:** Many retail brokers don’t expose real DOM (only synthetic 5-level). Verify with broker first or this filter does nothing. - **Honesty note:** Real value only on brokers with genuine ECN/STP DOM data (Pepperstone Razor, IC Markets Raw, FxPro Edge). On synthetic DOM brokers, this filter is theatre.

A.1.5 — Multi-Timeframe Footprint Stack - Mechanic: Currently footprint is checked only on entry timeframe (M5). Require footprint signal on entry TF *and* one HTF (M15 or H1). Two timeframes confirming = institutional commitment. - **Estimated contribution:** +2 percentage points - **Trade volume side effect:** -15 to -20% (HTF confirmation is rarer) - **Implementation:** Run footprint analyser on M15 in parallel; require both signals same direction within 3 bars. New input `RequireMTFFootprint: bool default false`. - **Honesty note:** This is genuinely strong because it filters out single-bar anomalies. The cost is real — your trade count drops noticeably.

A.1.6 — Correlated-Pair Confirmation (SMT++) - Mechanic: Already in Setup 20. Upgrade to a cross-cutting filter: every long EURUSD entry must show GBPUSD or DXY agreement; every short must show divergence. Strengthens *all* setups, not just Setup 20. - **Estimated contribution:** +1 to +2 percentage points - **Trade volume side effect:** -10% (occasional disagreement filters

legitimate setups) - **Implementation:** Subscribe to second symbol (`SymbolSelect()` MT5, `MarketData.GetSeries()` cAlgo). Compute a 20-bar correlation. New input `RequireCorrelationConfirm: bool default false` and `CorrelationSymbol: string default "GBPUSD"`. - **Honesty note:** Correlations break in news events. Build a guard: if rolling 20-bar correlation is below 0.5, disable this filter automatically.

Tier 3 — Real but small effect, complex to build

A.1.7 — Time-of-Day Bucket Filtering - Mechanic: Within Kill Zones, certain 30-minute windows historically outperform. After 200+ logged trades, filter out the bottom-quartile time buckets. - **Estimated contribution:** +1 to +2 percentage points (only after data accumulation) - **Trade volume side effect:** -10% - **Implementation:** Module reads `GODMODE_OFEA_log.csv` weekly, computes win rate by 30-min bucket, writes `time_bucket_blacklist.json`. EA reads the JSON and refuses entries in blacklisted buckets. New input `EnableAdaptiveTimeFilter: bool default false`. **Disabled until 100+ logged trades exist.** - **Honesty note:** Genuinely useful, but only after Phase 1 data exists. Don't enable on day 1.

A.1.8 — Volatility Regime Filter (ATR Z-score gate) - Mechanic: Compute ATR(14) Z-score over last 100 bars. Skip entries when $Z > +2.0$ (volatility spike — chaotic conditions, wider whipsaws) and when $Z < -1.5$ (dead market — no follow-through). - **Estimated contribution:** +1 to +2 percentage points - **Trade volume side effect:** -5 to -8% - **Implementation:** Compute in `RiskManager.cs`. New inputs `MaxVolZ: double default 2.0`, `MinVolZ: double default -1.5`, `EnableVolRegimeFilter: bool default false`. - **Honesty note:** The “skip extremes” intuition is correct but the thresholds are pair-specific. Tune per symbol.

A.1.9 — Asymmetric R:R Targeting - Mechanic: Currently SL is 1-2 ticks beyond aggression candle, TP is to POC. Upgrade: take only setups where TP1 (50% close) is at minimum 1.5R. Setups where TP1 is below 1.5R are skipped, even if all gates pass. This raises win rate because the loose-R setups are the ones that round-trip. - **Estimated contribution:** +1 to +2 percentage points (and +R) - **Trade volume side effect:** -10 to -15% - **Implementation:** New input `MinTP1_R: double default 1.5`. Computed before entry, skip if not met. - **Honesty note:** Genuine effect — the worst trades in any system are the “marginal R” ones. Cutting them helps both win rate and average R simultaneously.

Tier 4 — Speculative, treat with caution

A.1.10 — News-Surprise Avoidance Window - Mechanic: Currently news blackout = 15 min before/after scheduled releases. Extend to 30 min before/after, and add an unscheduled-volatility detector (sudden ATR spike $>3.0 Z$ without scheduled news = unscheduled headline). Skip for 20 min after. - **Estimated contribution:** +1 percentage point - **Trade volume side effect:** -5% - **Implementation:** Extend `NewsBlackout` checker. New input `NewsBlackout_Mins: int default 30`. Add unscheduled-vol detector to `RiskManager`. - **Honesty note:** Useful insurance but small effect on properly filtered setups.

A.1.11 — Streak-Aware Sizing (semi-Kelly) - Mechanic: This is *not* a win-rate booster — it's an EV booster that *looks* like one. After 2 consecutive winners, increase size by 25%. After 1 loss, return to base. After 2 consecutive losses, halve size. The streak-aware sizing protects equity during drawdown clusters. - **Estimated contribution: 0 percentage points to win rate** (be honest with yourself). +5 to +10% to risk-adjusted returns. - **Trade volume side effect:** ~0 - **Implementation:** New input `EnableStreakSizing: bool default false`. Track streak in `RiskManager`. **CRITICAL:** This does *not* help win rate. It helps *survivorship*. Don't conflate the two. - **Honesty note:** Including this only because operators routinely confuse “feels like higher win rate” with “actually higher win rate”. It's a sizing rule, not a filter.

A.1.12 — Post-Trade Adaptive Threshold Tuning - Mechanic: Every 50 logged trades, run a script that finds the optimal `VolZThreshold`, `MinAbsorptionStars`, and `LocTolATR` for the next 50-trade window. Adapts to regime drift. - **Estimated contribution:** +1 to +3 percentage points (regime-dependent) - **Trade volume side effect:** ~0 to -10%, varies - **Implementation:** External Python script reads `GODMODE_OFEA_log.csv`, runs grid search, writes `adaptive_thresholds.json`. EA reads on `OnInit` and on date change. New input `EnableAdaptiveTuning: bool default false`. **Disabled until 200+ logged trades.** - **Honesty note:** Single biggest source of long-term edge but also single biggest overfitting risk. Use walk-forward validation, not in-sample optimisation. If the script fits a regime that's already dying, you're worse off.

A.2 — STACKED EXPECTATION (HONEST MATH)

If you implement Tiers 1 and 2 fully (techniques A.1.1 through A.1.6), and they perform at the *midpoint* of their estimated contributions:

Base win rate:	75% (midpoint of 72-78%)
+ A.1.1 Quad Confluence:	+2.5
+ A.1.2 Liquidity Sweep:	+2.0
+ A.1.3 Round Number Bonus:	+1.5
+ A.1.4 DOM Confirmation:	+2.5 (broker-dependent)
+ A.1.5 MTF Footprint:	+2.0
+ A.1.6 Correlated Pair:	+1.5
Theoretical stack:	87.0%

That number is misleading and you must not believe it. Reasons:

1. **Filter contributions are not additive.** They overlap. The trades A.1.1 catches are largely the same trades A.1.5 catches.
2. **Each filter cuts trade volume.** Stacking all six cuts your daily count from 5-10 to maybe 1-3, contradicting your stated requirement.
3. **Diminishing returns are real.** The marginal trade you're filtering out at win rate 78% is closer to coin-flip — removing it boosts the *displayed* rate but only by a tiny amount in absolute terms.

Realistic stacked expectation when holding trade count near 5-10/day: - Pick **two from Tier 1 + one from Tier 2** =

realistic +4 to +6 percentage points - Hold trade count at 5-8/day instead of 5-10/day (small compression) - Result: **78-82% win rate range**, with 80% as the realistic midpoint of the new band

That is the honest answer to your question. **Targeting “80%+ at the same trade volume” lands in 78-82% —** call it 80% nominal — and the path is two Tier 1 filters plus one Tier 2 filter, not all twelve.

A.3 — RECOMMENDED IMPLEMENTATION SEQUENCE

Add these as new opt-in inputs to Section 7 of the main brief. Do **not** enable any by default — they earn their place through your own logged data.

Phase 9 (after the base build is verified, weeks 4-6): 1. Implement A.1.1 Quad Confluence (Wyckoff phase gate) — biggest single contributor 2. Implement A.1.9 Asymmetric R:R Targeting (`MinTP1_R = 1.5`) — improves both win rate AND R 3. Implement A.1.3 Round-Number Bonus — cheap, additive, low cost

Run these for 100 trades. Measure in your CSV log. If logged win rate sits at 78%+, you’ve earned the 80% band. If it sits at 75%, the regime isn’t paying for these filters and you stop adding more.

Phase 10 (only if Phase 9 produces +3% or better): 4. Implement A.1.5 MTF Footprint Stack 5. Implement A.1.2 Liquidity Sweep precondition (reversal setups only)

Run another 100 trades. Same measurement protocol.

Phase 11 (only after 200 logged trades, week 12+): 6. Implement A.1.7 Time-of-Day Bucket Filter — adaptive 7. Implement A.1.12 Post-Trade Adaptive Threshold Tuning — adaptive

Skip A.1.4 (DOM) unless you have confirmed your broker exposes genuine DOM. Skip A.1.11 (streak sizing) — it’s a sizing rule, not a win-rate filter; if you want to use it, use it for risk management, not to claim a higher win rate.

A.4 — NEW INPUT PARAMETERS TO ADD TO SECTION 7

```
=== 80% UPGRADE FILTERS (PHASE 9-11, ALL DEFAULT FALSE) ===
RequireQuadConfluence      : bool   default false // A.1.1 — Wyckoff phase gate
RequireLiquiditySweep_Rev  : bool   default false // A.1.2 — sweep before reversal
PrioritiseRoundNumbers     : bool   default true  // A.1.3 — score bonus only
RoundNumberTol_Pips       : double  default 3.0
RequireDOMConfirmation     : bool   default false // A.1.4 — broker-dependent
RequireMTFFootprint        : bool   default false // A.1.5 — needs HTF agreement
MTFFootprint_HTF           : enum {M15,H1} default M15
RequireCorrelationConfirm  : bool   default false // A.1.6 — SMT-style cross-pair
CorrelationSymbol          : string  default "GBPUSD"
EnableAdaptiveTimeFilter   : bool   default false // A.1.7 — needs 100+ trades
MaxVolZ                    : double  default 2.0  // A.1.8 — vol regime
MinVolZ                    : double  default -1.5
EnableVolRegimeFilter      : bool   default false
MinTP1_R                   : double  default 1.5  // A.1.9 — asymmetric R:R
NewsBlackout_Mins          : int     default 15   // base 15, push to 30 for A.1.10
EnableUnscheduledVolDetect : bool   default false // A.1.10
EnableStreakSizing         : bool   default false // A.1.11 — sizing only
EnableAdaptiveTuning       : bool   default false // A.1.12 — needs 200+ trades
```

A.5 — NEW NOTIFICATION TAGS FOR THE OPERATOR DASHBOARD

Add to the cascade in Section 4 of the main brief:

Tag	Meaning	Visual	Sound
N-M	Quad confluence achieved (Wyckoff phase confirmed)	Gold-bordered diamond + “WYCKOFF Phase D” badge	special chime
N-N	Liquidity sweep detected pre-reversal	Lightning-bolt icon at swept level + “SWEEP↓” or “SWEEP↑”	none
N-O	Round-number proximity bonus	Small “★RN” badge on price tag	none
N-P	DOM imbalance confirmed	Stacked bid/ask block visual + “DOM 2.3:1 BID”	single beep
N-Q	MTF footprint agreement	Two-bar timeline icon + “M5+M15 ✓”	single beep
N-R	Correlated pair confirms	Linked-pairs icon + “GU ALIGN” or “GU CONFLICT”	none
N-S	Vol regime gate triggered	“VOL Z=2.4 — SKIP” red banner	warning beep
N-T	Time-bucket blacklisted	“BUCKET FILTER — SKIP” amber banner	none
N-U	Adaptive thresholds reloaded	Dashboard footer flashes “ADAPT v3 — N=247”	none

Update the dashboard layout in Section 3 to show **rows 9 through 12** when Phase 9+ filters are enabled — additional checklist items above the score line.

A.6 — THE OPERATOR’S HONESTY CONTRACT

Sign off on these statements before enabling any A.1.x filter:

- I will measure each filter’s contribution from my own CSV log** — not trust the estimates above. Estimates are starting points; my data is truth.
- I will not enable a filter just because it sounds powerful.** I will enable one filter at a time and measure 50-100 trades before adding the next.
- I will accept that 80%+ may not be achievable on my pair / my broker / my regime.** If the data refuses to cross 78%, I will stop adding filters and accept the actual edge.
- I will not confuse small samples with proof.** A run of 15 winning trades is noise. 100 trades is data. 300 trades is conviction.
- I will not chase win rate at the expense of R:R.** A 90% system with average +0.3R is worse than a 70% system with average +1.5R. The objective is *expectancy*, not headline win rate.

A.7 — FINAL BUILD ORDER UPDATE

In Section 17 of the main brief, **insert these phases** after Phase 8:

PHASE 9 — Tier 1 Win-Rate Boosters (after 100 base-system trades logged) 28. Implement A.1.1 Quad Confluence (Wyckoff phase gate in `OF_WyckoffPhase.mqh` / `WyckoffPhase.cs`) 29. Implement A.1.9 Asymmetric R:R Targeting in `RiskManager` 30. Implement A.1.3 Round-Number Bonus scoring in `SetupDetectors` 31. Add notifications N-M, N-N, N-O to NotificationCenter 32. Add dashboard rows 9, 10, 11 to `Dashboard` 33. Run 100 trades. Measure delta. Stop if delta < +2pp.

PHASE 10 — Tier 2 Boosters (only if Phase 9 hits +3pp) 34. Implement A.1.5 MTF Footprint stack 35. Implement A.1.2 Liquidity Sweep precondition (`LiquidityScanner` module) 36. Add notifications N-Q, N-N 37. Run another 100 trades.

PHASE 11 — Adaptive Layer (only after 200+ total trades logged) 38. Build external Python analyser for time-bucket and threshold optimisation 39. Implement A.1.7 and A.1.12 with JSON-driven inputs 40. Add notifications N-T, N-U 41. Walk-forward validation before deploying any adaptive output

A.8 — STANDING ORDER FOR APPENDIX A

The base brief delivers a 72-78% system. This appendix delivers a path to 78-82%, honestly framed as “the 80% band” rather than the impossible “consistent 85%+”. Implement the filters as opt-in Phase 9-11 work, measure every one against your own logged trades, and stop adding filters the moment your CSV proves diminishing returns. The discipline of not adding more is itself part of the edge.

Appendix A authored against the same source authority as the base brief. No new external sources introduced. All estimated contributions are honest mid-points; operator is contracted to verify against logged data before treating any estimate as fact.